

The Global Drifter Program hourly dataset

DATA3001 DATA SCIENCE AND DECISIONS IN PRACTICE TERM 3, 2026

1. DATA CONTEXT

The upper few metres of the ocean are where it exchanges heat, moisture and carbon with the atmosphere above. This layer also transports material such as pollutants, oil, plastic, larvae, and objects lost from vessels. Search and rescue, spill response and fisheries management all rest on the same estimate of where the surface water goes, how fast, and how well that is known. In general, measurements of ocean quantities are difficult to obtain. The ocean is large, and at any moment very little of it is under observation; what is known about surface currents comes from a small number of observing platforms, each of which trades capability against cost.

- *Satellites* cover nearly the whole ocean every few days. The instrument is expensive to build and launch, but once it is in orbit each additional observation costs almost nothing. They do not measure current at all. Velocity is inferred from the height and temperature of the sea surface; the resulting fields are smooth in space and slow in time, features tighter than about a hundred kilometres are lost, and the retrievals still have to be calibrated against measurements made in the water.
- *Moorings* are anchored to the sea floor and measure the passing flow directly, at high temporal resolution and over periods of years. Each one costs of order hundreds of thousands of dollars to build, deploy and maintain, and reports a single location.
- *Ships and gliders* can be sent to a chosen place, which suits a specific question in a specific place. Ship time runs to tens of thousands of dollars a day, so the records are sparse.
- *Drifters* are expendable buoys, released and never recovered. They cost a few thousand dollars each, measure the current directly, and report for one to two years. They cannot be steered: a drifter samples wherever the flow carries it.

That drifters sample along the flow is a feature of the design. A mooring records one fixed point continuously for years. The drifter array covers a far larger area, but sparsely, and at positions as a function of the flow, rather than at chosen locations. This is the classic distinction between an Eulerian description, fixed in space with the water passing through it, and a Lagrangian one, which follows a parcel of water. The Global Drifter Program is the largest Lagrangian dataset in the world.

Each drifter is a simple instrument. A surface float carries a satellite transmitter and a thermistor for sea-surface temperature; below it, on a long tether, hangs a holey-sock, or *drogue*, centred at fifteen metres depth. It acts as a sea anchor: the buoy follows the water at that depth instead of being pushed across the surface by wind and waves. A drogued drifter tracks the current to within about a centimetre per second; when a drifter loses its drogue (and most eventually do) its recorded velocity is contaminated by wind slip and reads systematically faster. Drogue status is recorded per buoy.

The array is maintained thanks to coordinated efforts by WMO and IOC-UNESCO (see <https://www.ocean-ops.org/dbcp/>). Drifters supply most of the in-situ sea-surface temperature against which satellite retrievals are calibrated, and the mean currents estimated from them are the reference against which ocean circulation models are assessed.

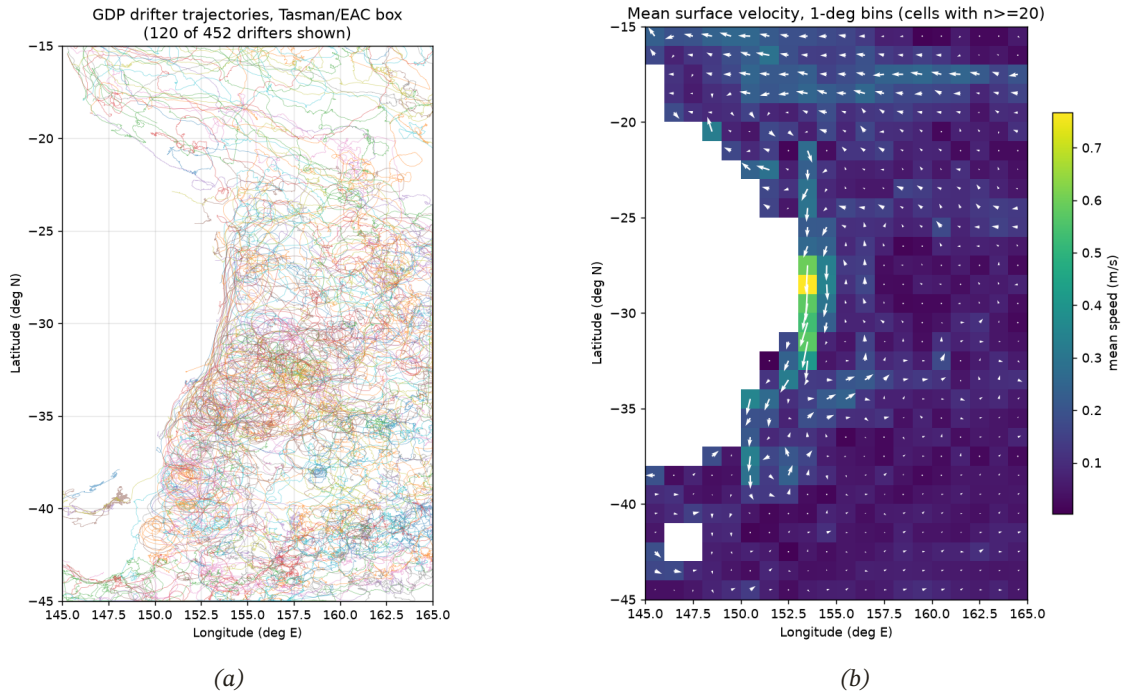


Fig. 1. The drifter records, and the field estimated from them. (a) Tracks of 120 of the 452 drifters that have passed through the worked-example region off eastern Australia; each line is one buoy. The coastline appears as a void; the tracks concentrate along the boundary current and pool in the eddy field to the south. (b) The same records averaged onto a one-degree grid, over cells containing at least twenty observations. The East Australian Current appears as a southward jet along 153°E, strongest near 30°S, separating from the coast and turning east near 34°S.

2. THE GLOBAL DRIFTER PROGRAM HOURLY PRODUCT

The Global Drifter Program is part of, and the main component of, the Global Surface Drifter Array which is an international collaboration (including Australia!). The US component forms NOAA's operational array of satellite-tracked surface buoys, run jointly by the Atlantic Oceanographic and Meteorological Laboratory in Miami and the Lagrangian Drifter Laboratory at Scripps. The design target is a global array of about 1,300 drifters on a nominal $5^\circ \times 5^\circ$ spacing; at the time of writing 1,436 are reporting. Roughly 25,000 drifters have been released since the array began in the late 1980s, and each contributes its track until it ceases to transmit or comes ashore.

Typically, drifters are located by satellite, and the raw fixes arrive irregularly and with error (some are also equipped with their own GPS locators). The Argos system, which has tracked most of the array, gives positions with errors of order a hundred metres at an average interval of 1.2 hours; GPS-tracked drifters are located far more precisely. The sampling is uneven and the position errors are not negligible. Velocity is not measured at all: it has to be derived from successive positions, and differencing a noisy, irregularly spaced track can be awkward.

The hourly GDP product exists as a dataset cleaned for scientific use. Elipot et al. (2016) selected an interpolation scheme by testing candidates against 82 drifters that were tracked simultaneously by Argos and by GPS in the North Atlantic, using the GPS tracks as ground truth; the chosen method maps the irregular fixes onto a uniform one-hour grid and returns position, eastward and northward velocity, and an uncertainty for every estimate. The sea-surface temperature component was built on the same grid and documented separately (Elipot et al., 2022).

The hourly grid, in place of the six-hourly product that preceded it, is what makes the dataset useful at a regional scale. Sampling every six hours puts the Nyquist period at twelve hours, which sits

directly on the semidiurnal tide at 12.42 hours and on the inertial oscillation at high latitudes. Those motions do not simply disappear at six-hourly resolution; they are aliased into lower frequencies, where they contaminate whatever else is being estimated. An hourly grid moves the Nyquist period to two hours and puts both bands safely inside the resolved range.

The product is available in a AWS S3 bucket (and may be accessed with `clouddrift` from python):

- *The AWS Open Data registry* holds a Zarr store in the public bucket `noaa-oar-hourly-gdp-pds`, readable lazily so that only the chunks you touch cross the network.
- *CloudDrift* (<https://clouddrift.org>), a Python package written by the authors of the dataset, wraps that store: `from clouddrift.datasets import gdp1h` and then `ds = gdp1h()` hands back the whole product as a lazy xarray object.

3. PROJECTS

3.1. Regions

You will be responsible for selecting a region of interest R . It is important that this is well motivated as an important global current, area of economic interest, or similar. For instance, the box in Fig. 1 is the East Australian Current off New South Wales, which we use throughout as the worked example (do not choose this region). You are responsible for assessing if your region has enough data-points available. A region in the middle of the South Pacific may hold only a few thousand observations, too thin for a gridded climatology. Nothing in your code should depend on the exact R , and any functionalities should work if given a different region.

3.2. Viable projects

Here are five viable projects, given the drifter data. This is an open ended assignment and so you may extend past these ideas in your analysis of the data and the delivery of a product. Begin by selecting a main project, and develop it alongside your group and with your lecturer.

3.2.1. The mean circulation of R

What is the time-averaged surface current in R , and how sure can you be of it in any given part of the map? Where does it reverse with the season? The drifters do not sample R evenly. Some cells rest on thousands of observations and some on a handful. Worse, their positions are set by the flow you are trying to estimate: they gather where the surface water converges and are swept out of where it diverges, so a cell that is thick with observations may be thick with them for reasons that bias what you compute there. In and off itself, an unweighted mean of the velocities in a cell is not, in general, a useful estimate of the mean current. The same question applies to sea-surface temperature.

Product. A gridded climatology of R that another analyst could load and use. It should carry the mean flow and the seasonal fields, and it should provide a cell by cell metric of uncertainty.

3.2.2. Where things end up

Something goes into the water at a chosen point in R : a container off a ship, a life raft, a slick of oil. Where has the surface flow taken it a week later? A month later? A year? At the scale of the whole box, which parts of R feed which others, and which are cut off from the rest?

Product. A surface-transport operator for R. E.g. a transition matrix (or similar) someone else could pick up and iterate forward, together with what it says about where a few chosen release points end up, and about where R as a whole gathers material and where it loses it.

3.2.3. Whether the record can support a trend

Can the data in your region support evidence for a trend you might report in current speed or surface temperature? Has the sampling of R changed over the thirty-five-year record and how does this affect your confidence in a decades-long trend? The array grew from a scattering of buoys to roughly thirteen hundred, and the tracking technology changed from Argos to GPS partway through; on top of that, deployments follow the shipping and the research cruises. An apparent change in what the data say about R may therefore be a change in R, a change in who was measuring it, or both. A well-established negative result is a perfectly good answer here.

Product. A sampling history of R, and the multiple trend estimates: e.g. once naively, once with the change in sampling accounted for. Finish with a verdict on whether the record for R can carry any claims of trends.

3.2.4. Predicting where a drifter goes next

Given a drifter's recent track, where will it be in an hour's time, or a day's, or a week's? There are two obvious baselines to beat: that the drifter carries on at its present velocity, and that it follows the mean flow. This differs from project 3.2.2. that it concerns the near-time prediction of a *single* drifter, not the mean transport properties of the flow field.

Product. A forecast evaluation on held-out drifters. Put error against lead time for your model and baselines on the same axes. State test-training split and model development.

3.2.5. Identifying extremes

Could you tell, from a drifter's record *in real-time*, that it is sitting in a marine heat wave or an unusually strong current? With hindsight, the extremes in R can be found easily; the question here is whether a detector running on the incoming data would have fired at the time, early enough to be useful. An extreme is defined against a normal-state. The normal varies with position and season, so you need a baseline for a place the drifter may never have been. A rise in recorded temperature may mean the water warmed, or it may mean the drifter travelled somewhere warmer. Similarly, a jump in speed may be an eddy, or the drifter has entered a predictably energetic region.

Product. A detector that runs on a drifter's record as it arrives, with a verified false-alarm rate and a detection delay, validated by replaying historical drifters through it as though the observations were coming in live.

SOURCES

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